



Seminar on - Climate Change : Mitigation Strategies in Agriculture & Energy Sectors December 19-20, 2017

THE SEMINAR

This two day Seminar cum Round Table on "Climate Change: Mitigation Strategies in Agriculture & Energy Sectors" will be held at Ram Krishna Dharmarth Foundation (RKDF) University on 19-20, Dec 2017 at our Gandhi Nagar campus, near RGPV, Bhopal. This seminar is 'Sponsored' by EPCO, Bhopal and is being organized to put the technology transfer and development issues in perspective and take stock of the R&D efforts of various Stakeholders in Climate Change Mitigation in both the major sectors viz. Energy & Agriculture. The knowledge partners apart from EPCO will be MPUVNL, CPRI, BHEL, IEL, RPI- USA, TOSHIBA Japan and Institutes under aegis of AUAP-Thailand. The comprehensive conference program will include key note addresses from national experts of Agricultural and Energy discipline as well as our Project- collaborator expert from RPI, New York, USA.

:: Key Speakers ::

Dr. Panjab Singh (President NAAS, New Delhi & Advisor RKDF University)

Dr. Anil Kumar Singh (VC RVSKVV, Gwalior & Ex-Deputy DG ICAR)

Prof. Partha S Dutta (Dy. Director, RPI, New York, USA)

Dr. Ashok K. Patra (Director Indian Institute of Soil Sciences)

Dr. K. K. Singh (Director CIAE, Bhopal)

Dr. P. K. Agarwal (National Professor Environmental Sciences, IARI)

Dr. V. S. Tomar (Vice Chancellor JNKVV, Jabalpur)

Chief Patron: Dr. Sadhna Kapoor (Chancellor, RKDF University)

Patron: Mr. Siddharth Kapoor (MD, RKDF University)

Chairperson: Dr. Panjab Singh (President NAAS, New Delhi & Advisor RKDF University)

Convener: Dr. V. K. Sethi (Vice Chancellor, RKDF University)

Organizing Secretary: Dr. B. N. Singh (Registrar, RKDF University)

:: Organizing Team ::

Dr. Shashi Singh (Principal Agriculture Department, RKDF University)

Dr. Amit Pandey (Deputy Registrar, Academics)

Mr. Aditya Gupta (Director Training & Placement, RKDF University)

Mr. Nimish Singh (TA, RKDF University)

Mr. Saurabh Kumar (TA, RKDF University)



SEMINAR ON CLIMATE CHANGE : MITIGATION STRATEGIES IN AGRICULTURE & ENERGY SECTORS

(Sponsored by EPCO, Bhopal)



December 19-20, 2017

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CLIMATE CHANGE: A GLOBAL CHALLENGE

The Climate change caused by the excessive industrial emissions of greenhouse gases (GHGs) is one of the gravest challenges being faced by our planet today. Studies have shown that there is no single option for combating this problem, but rather, a portfolio of measures in various sectors of economy like Energy, Agriculture, Urban & Rural Habitat and various measures related to environment protection and ecological sustenance. The whole world is aligned today for Mitigating Climate Change. Educational Institutes & Universities around the globe are involved in giving their best for saving the mother earth.

Conferences of Parties (COPs) under aegis of UNFCCC are the most effective International efforts in framing policies for mitigation of Climate change. COP 21 at Paris and its Declaration 2015 has been a landmark in which consensus was achieved to limit global warming "well below 2 degree C". The Global Warming to date has reached +1.01383959697 degree Centigrade and the time left to reach 2 degree C is estimated to 29 years, 2 months and 1 day with the current rate of global emission of 40,000 Million Tons per annum (MTPA).

Energy & Agriculture (inclusive of Land Use) are the two major sectors of economy which are responsible for over 80% of total GHG emission in India. At the same time Agriculture production is the one that is most vulnerable to the Climate change that is taking place globally and particularly in developing world.

CLIMATE CHANGE: STRATEGIES IN ENERGY SECTOR

Man's expanding need for energy has become almost synonymous to the growth and developments in the modern S&T propelled agro-industrial society. It is however quite clear that increased energy consumption has resulted into serious problems such as global warming and erratic climate changes currently being experienced around the world. Our Power sector, mainly the Coal based Conventional Power plants emit almost 56% of our total CO₂ Emission which is at the level of about 2170 Million Tons Per Annum (world total emission being at the level of above 40,000 MTPA).

In the Energy sector, India is committed to pursue Green Power Mission. It has declared to reduce its GHG intensity by 33-35% by 2030, through installation of 175 GW Renewable Power (100 GW Solar) by 2022 which will be 40% of our total Power Generating Capacity at that time (today's level being 3, 29, 231 MW). Developmental process using Coal will continue, however Carbon intensity to be cut by 33%, not carbon emission in absolute terms. Countries between Tropic of Cancer & Tropic of Capricorn have already formed International Solar Alliance, a consortium to promote Solar, in which India is in lead role.

The recently adopted Nationally Determined Contributions (NDCs) as signatory to the United Nations Framework Convention on Climate Change (UNFCCC) emphasize the importance of demand side factors. In its submission, India gave particular importance to the behavioral change leading to energy conservation, something that has received insufficient attention in the developing countries. The National Action Plan aims to adopt this shift in our energy policy.

CLIMATE CHANGE: STRATEGIES IN AGRICULTURAL SECTOR

India has a diverse physiography surrounded by oceans along its southern borders, Himalayas at its north a vast plateau at its center, desert on its western region and low rising hills on its south eastern and south western borders resulting in a climate that is highly variable across its expanse. Here Still around 60% population is dependent on agriculture and 40% of agricultural land is rain-fed making agriculture dependent livelihoods highly vulnerable to climate Change. This vulnerability may in due course pose a threat to the food security of the Nation therefore climate sensitive sectors like forestry water resources urban and rural development human and animal health etc. are equally vulnerable. In addition an unequal social and economic development across the country had resulted in differential coping capacities amongst the population. The changing climatic parameters are likely to accelerate the vulnerabilities further; unless actions are taken now to adapt to same. The Climate change impact on Agriculture leads to over 1.5% loss in India's GDP.

Noteworthy interventions to reduce adverse impacts of climate change on Agriculture being pursued in Mission mode include improvement in forecasting & early warning systems, establishing hazard & vulnerability mapping, augmenting public awareness, creating community-based forest management and afforestation projects and Improvement in irrigation through various schemes in mission mode:

- National Food Security Mission (NFSM)
- National Food Security Mission-Commercial Crops
- Mission for Integrated Development of Horticulture (MIDH)
- National Mission on Oilseeds and Oil Palm
- National Mission for Sustainable Agriculture
- National Mission on Agricultural Extension and Technology

RKDF UNIVERSITY AT THE INTERNATIONAL CLIMATE CHANGE FORUM

The RKDF University, one of the four universities of RKDF group, the largest educational group in central India; is fulfilling its commitment of Carbon foot print reduction and focusing on area of R&D in Green & Clean Technologies in Energy & Agriculture fields. We have set up 500kW Roof top Solar PV plants on our building roofs and so far saved over 2000 tons of CO₂ in past 3 years. Apart from this several Research projects are being pursued in Renewable Energy Technologies, Thermal Storage and Carbon Capture & Sequestration (CCS).

Our Chancellor Dr. Sadhna Kapoor as Honorary Professor of Oxford University has presented a key note address on 'Climate Change: Vulnerability studies in Energy & Agriculture disciplines in Indian Universities' at Lucerne Switzerland on 3rd July this year before the International Academic community. Also our Vice Chancellor Prof. V K Sethi has presented India's stand in CCS at the World Climate Conference - GHGT-13 at Lausanne from 14-18 November 2016 and was a member of Indian Delegation to COP 22 of UNFCCC. Also the University has marked its presence at the International Energy Agency's CCS meet at Birmingham Alabama from 5-8 Sept 2017 in which the VC of the University presented a key note address on "Small Steps by India to this Great World" in Adaptation & Mitigation of Climate change.